

The empirical recurrence for OEIS A283491, recovered from its tail

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Abstract

OEIS A283491 records “Empirical recurrence of order 74” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 218$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 78$. Its last coefficient is $c_{74} = -6561 \neq 0$, so no further tail satisfies anything shorter; any order-74 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A283491 is “Number of $n \times 5$ 0..1 arrays with no 1 equal to more than three of its horizontal, vertical and antidiagonal neighbors, with the exception of exactly one element.”. It has offset 1 and begins

0, 64, 5444, 174232, 5691512, 173139952, 5074155320, 145077026768, ...

The entry states:

Empirical recurrence of order 74 (see link above)

As of the “Last modified” line on the live entry (Mar 08 2017 14:13:59, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 218$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 218$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 74: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 74.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 436$ exact terms from $n = 4$ on, it returns order 74 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{74} c_i a(n-i),$$

c_1	26	c_{20}	−1821970547687	c_{39}	10058423319268472	c_{58}	502125419233
c_2	287	c_{21}	2941932024020	c_{40}	−9236622971609794	c_{59}	−71230311584
c_3	−3286	c_{22}	−1175634844479	c_{41}	8194062031950736	c_{60}	−42263539411
c_4	−79156	c_{23}	−12096295227726	c_{42}	−6959687459417963	c_{61}	41032471698
c_5	−682342	c_{24}	47455703073902	c_{43}	5517828249200984	c_{62}	−21163889490
c_6	−3543945	c_{25}	−105419166917404	c_{44}	−4112738466477452	c_{63}	9212123846
c_7	−12585958	c_{26}	165822289401798	c_{45}	2999949802811048	c_{64}	−3893282082
c_8	−33144499	c_{27}	−163176772374338	c_{46}	−2130091139354827	c_{65}	1517098126
c_9	−71591798	c_{28}	−5120763432641	c_{47}	1368288746525800	c_{66}	−506035228
c_{10}	−175519914	c_{29}	453495828643776	c_{48}	−758519579548088	c_{67}	150584094
c_{11}	−524515026	c_{30}	−1242100159605174	c_{49}	387075163013808	c_{68}	−44991936
c_{12}	−1431380680	c_{31}	2380243787559154	c_{50}	−200494073314535	c_{69}	13134708
c_{13}	−2272872486	c_{32}	−3807562374132294	c_{51}	89645510461900	c_{70}	−3298644
c_{14}	−4125109786	c_{33}	5367196165209188	c_{52}	−13939879086040	c_{71}	684936
c_{15}	−3462124924	c_{34}	−6967180680190321	c_{53}	−21007009697976	c_{72}	−143937
c_{16}	−27728861781	c_{35}	8467784644684784	c_{54}	22688916333933	c_{73}	36450
c_{17}	−7242049532	c_{36}	−9620185404930857	c_{55}	−12814291931488	c_{74}	−6561
c_{18}	−161642552108	c_{37}	10297484563855618	c_{56}	5180022028034		
c_{19}	582900374344	c_{38}	−10452876171416148	c_{57}	−1747625650976		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 78$, and it is the recurrence OEIS A283491 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{74} - c_1 x^{73} - \dots - c_{74}$. Its constant term is $-c_{74} = 6561$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 74 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 74, so $\deg q = 74$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 74, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 74, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -6561 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-74 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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